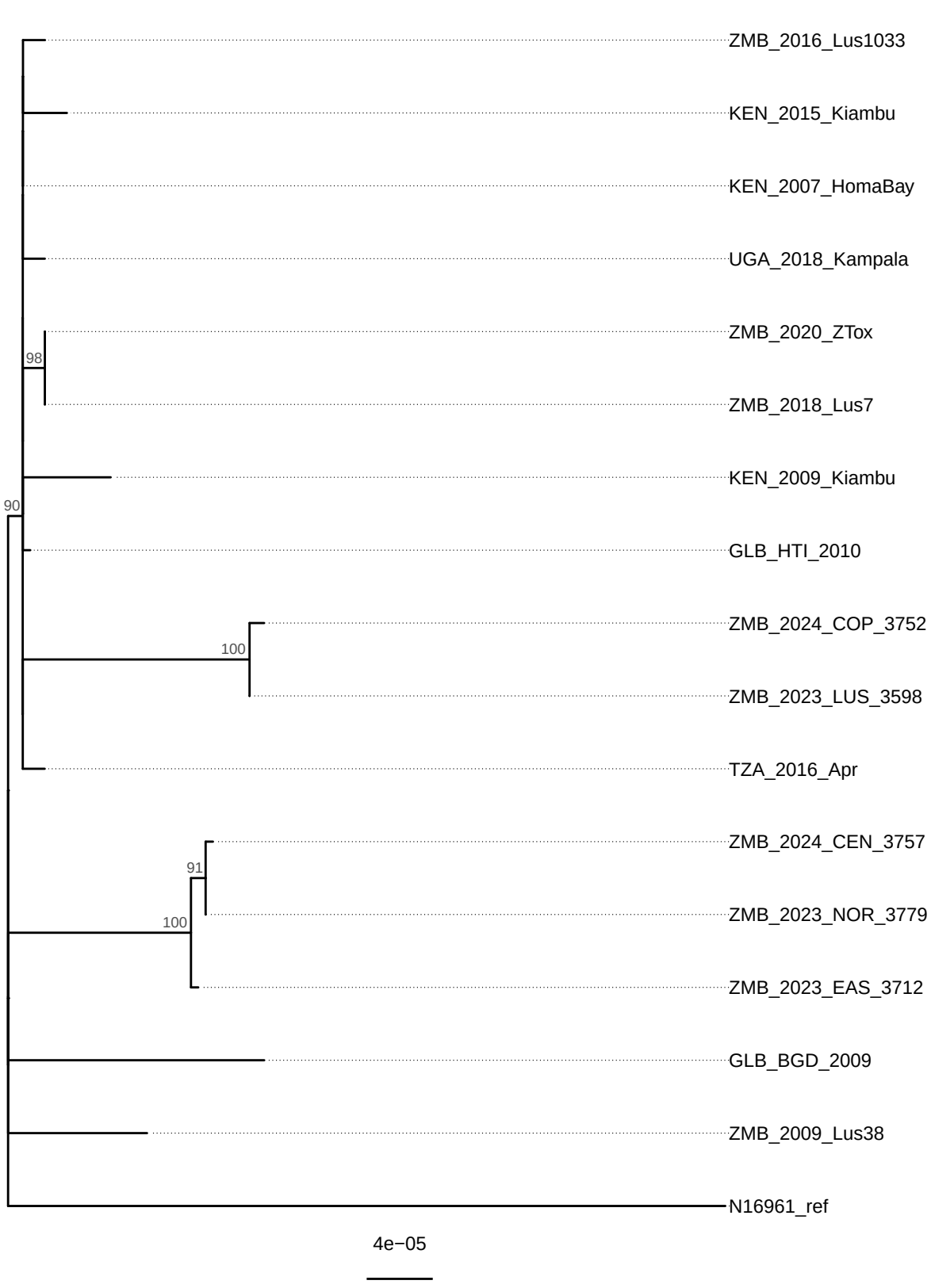
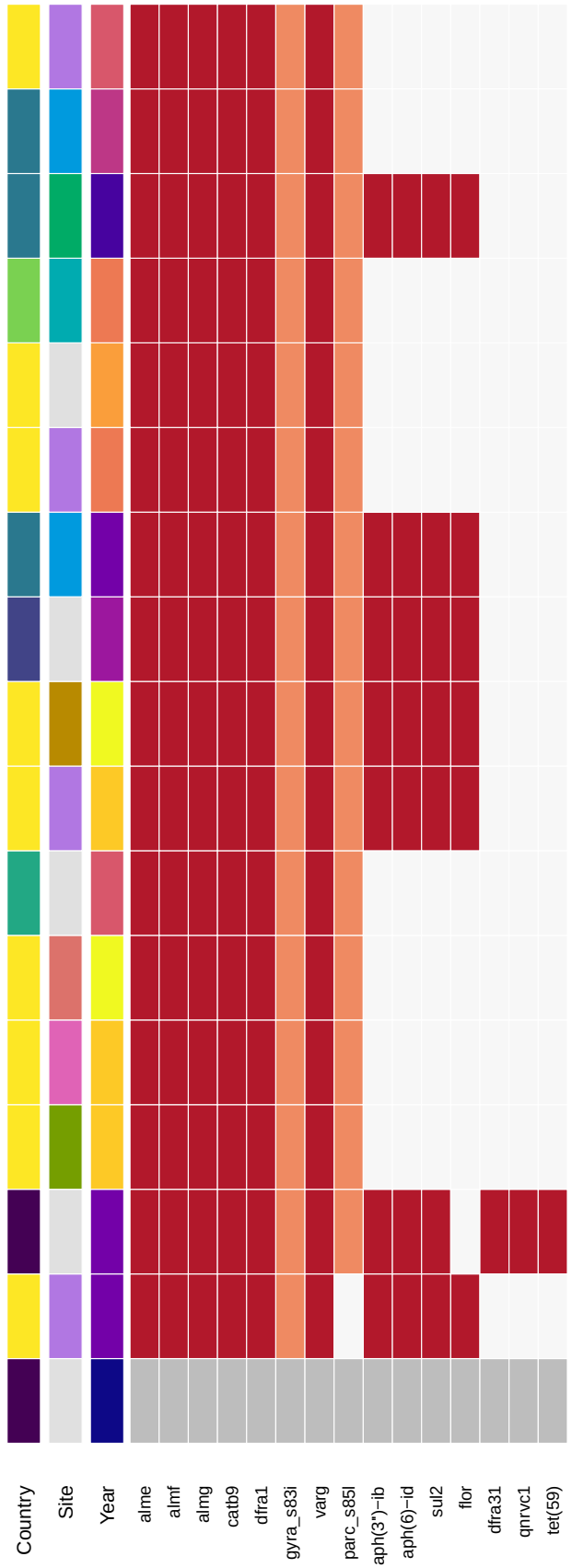
Core−genome phylogeny with resistance−associated determinants

Rooted on N16961_ref (reference genome; a pragmatic root, not a true outgroup).
Node labels: ultrafast bootstrap, shown when >= 70. Genotype only; not phenotype.





Cholera toxin genes (ctxA/ctxB) are virulence markers and are not shown. Gene columns with a numeric suffix were merged. Partial contig−end hits (blaI, blaZ) are not counted.